

GraphQL Blog — Data Flow Analysis

Project: GraphQL Blog

Technology: Python, FastAPI, Strawberry GraphQL, SQLAlchemy, PostgreSQL

Location: /home/dominic/Documents/graphql-blog

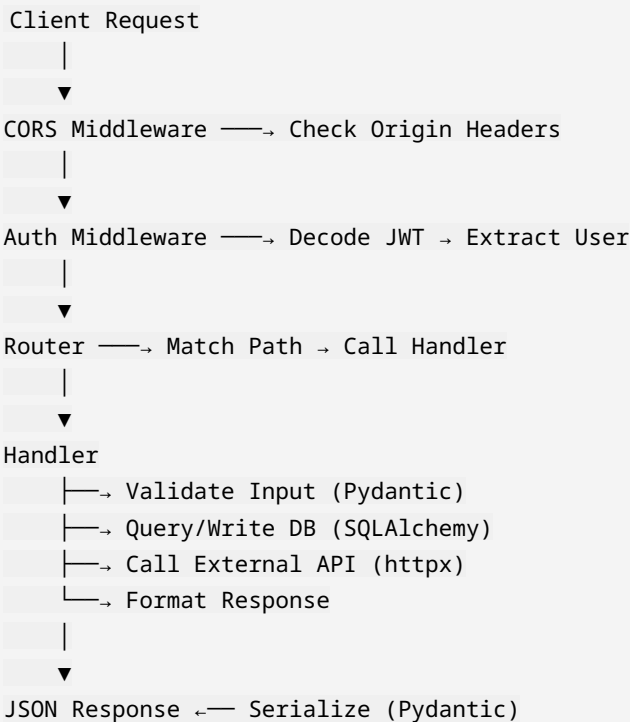
Report #: 15/20

Request Flow

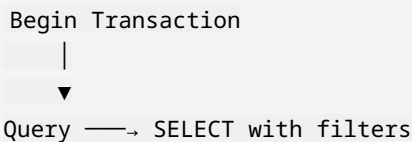
Client → GraphQL Query → Strawberry Resolver → SQLAlchemy → PostgreSQL → Response

Data Flow Analysis

Request Data Flow



Database Transaction Flow



```

|
▼
Validate —→ Check business rules
|
▼
Mutate —→ INSERT/UPDATE/DELETE
|
▼
Commit —→ On success
Rollback —→ On error

```

WebSocket Data Flow (Queue Module)

```

Client opens WebSocket
|
▼
Server accepts connection
|
▼
Task created → Server pushes progress updates
|
▼
Client receives: {"progress": 45, "status": "running"}
|
▼
Task completes → {"progress": 100, "status": "completed"}
|
▼
WebSocket closes

```

External API Data Flow (AI Chat Proxy)

```

Client → POST /v1/chat/completions
|
▼
Proxy → Add API key → Forward to api.openai.com
|
▼
OpenAI → SSE stream (tokens)
|
▼
Proxy → Forward stream as-is
|
▼
Client → Parse SSE → Render tokens incrementally

```

Data Transformations

1. **URL shortening:** Long URL → short_code (base62 hash)
2. **Auth:** Password → bcrypt hash → DB

3. **RAG:** PDF → Text → Chunks → TF-IDF vectors
4. **Queue:** Task request → DB row → WebSocket progress
5. **Chat:** Message → JSON → API call → SSE stream